Complex Analysis

If

$$x^2 + 1 = 0$$

Then x is?

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Complex analysis (traditionally known as the theory of functions of a complex variable) is the study of complex numbers together with their derivatives, manipulation, and other properties and investigates function of complex numbers. Complex analysis is an extremely powerful tool with an unexpectedly large number of practical applications to the solution of physical problems. Contour integration, for example, provides a method of computing difficult integrals by investigating the singularities of the function in regions of the complex plane near and between the limits of integration.

Why complex number system is introduced?

Mathematics has its own language in which Alphabets are numbers, so number system is so important in mathematics. The prince of mathematics Gauss said that every polynomial has at least one root and that Polynomial has maximum roots exactly equal to its order or degree. Above statement is called the fundamental theorem of Algebra. When we consider the polynomial equation like as $x^2+1=0$ that implies $x^2=-1$ but which is not possible in the real field because square of any real number is non-negative so real field fails to give the solution of this polynomial equation. But fundamental theorem of Algebra tells it has maximum two roots. To permit the solution of the equation $x^2+1=0$ and similar types, the set of complex numbers is introduced. Therefore, complex number system includes real number system as a subset, so complex number system is the extended form of real number system that solved the above considered problem. By considering $i^2=-1$ the problem $x^2+1=0$ provides two complex roots that covered the fundamental theorem of Algebra stated by great mathematician Gauss .It is notable that the mathematician Euler first use the symbol i for imaginary unit and its geometrical meaning in the complex plane is the point $(0,1)$. Solution of the above arises problem is as follows:

$$x^2+1=0 \qquad \Rightarrow x^2=-1$$

$$\Rightarrow x^2=i^2 \quad [\because i^2=-1]$$

$$\Rightarrow x=\pm\sqrt{i^2} \qquad \Rightarrow x=\pm i$$

Finally, we conclude that the roots of the aroused problem are $x=i$ and $x=-i$.

Complex variable:

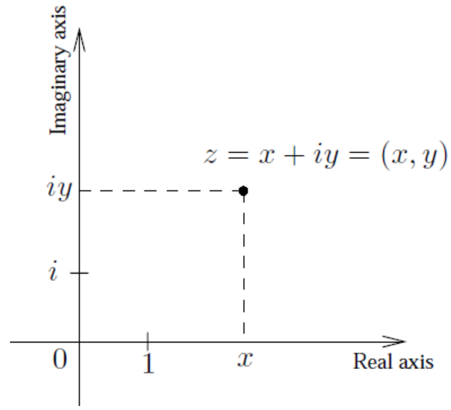
Complex variable z is the special type of linear combination of two real variables x and y with the special sign i that is $z=x+i\,y$ where $x\in R, y\in R$ *and* $i=\sqrt{-1}$.

In complex variable z, x is the real part of z denoted by the symbol Re (z) =x and y is the imaginary part of z denoted by the symbol Im (z) =y and also i is called imaginary unit. Geometrically complex variable represents general point in the complex plane/Argand Plane/Argand diagram/Gaussian Plane. Also Geometrically, Re(z) is the projection of z = (x, y) onto the x axis, and Im(z) is the projection of z onto the y axis. It makes sense, then, that the x axis is also called the real axis, and the y axis is called the imaginary axis.

Various form of Complex variable:

Cartesian form/Rectangular form:

The form of the complex variable $z=x+i\,y$ where $x\in R, y\in R$ *and* $i=\sqrt{-1}$ is called Cartesian form.



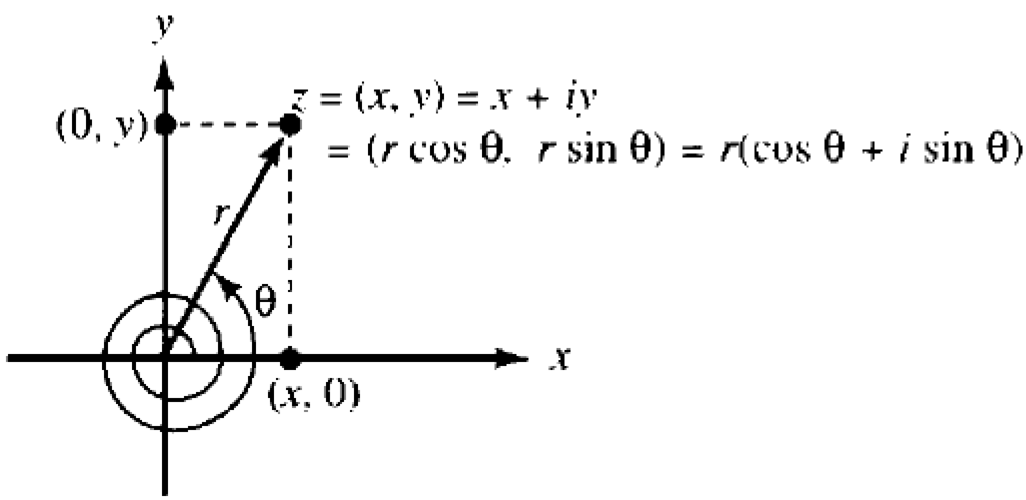
Polar Form:

We have Cartesian form as like as $z = x + i\,y$

From the relation of Cartesian and polar system we have $x = r\cos\theta$ and $x = r\cos\theta$.

Now,

$$z = x + i\,y$$
$$z = r\cos\theta + i\,r\sin\theta$$
$$z = r(\cos\theta + i\sin\theta)$$, this is the polar form of z.



Exponential Form:

We have Cartesian form as like as $z = x + i\,y$

From the relation of Cartesian and polar system we have $x = r\cos\theta$ and $x = r\cos\theta$.

Now,

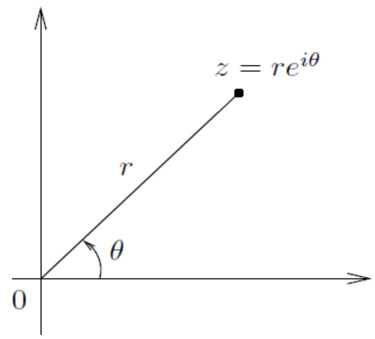
$$z = x + i\,y$$
$$z = r\cos\theta + i\,r\sin\theta$$
$$z = r(\cos\theta + i\sin\theta)$$

We have Euler formula

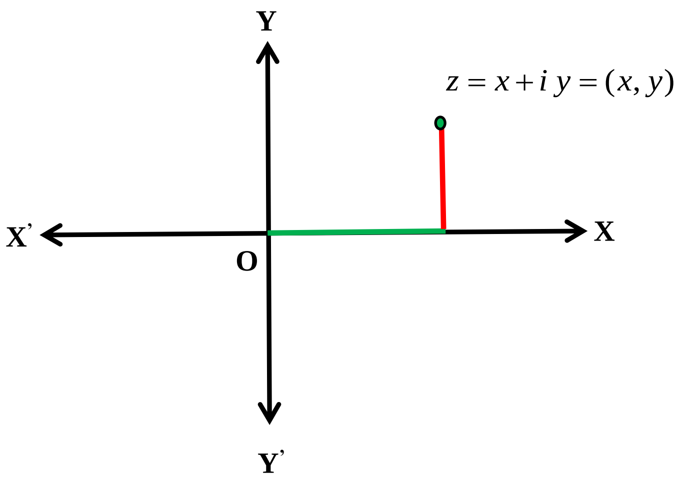
$$e^{i\theta} = \cos\theta + i\sin\theta = cis\theta$$

Here $z = r(\cos\theta + i\sin\theta)$

$z = re^{i\theta}$, this is the exponential form of z.

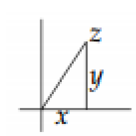
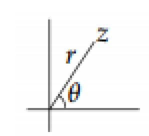


Complex Number:

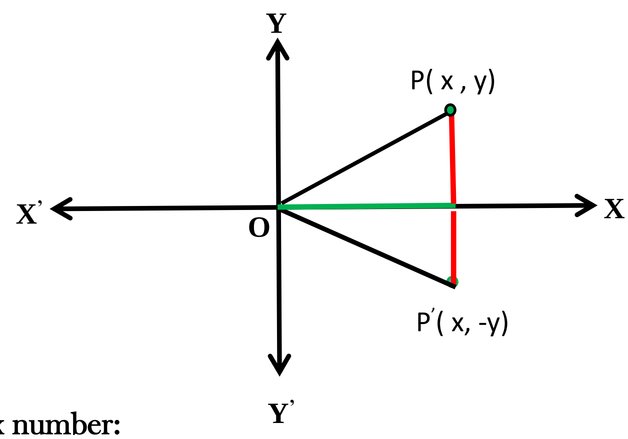


The complex variable $z = x + i\,y$ is said to be complex number if x and y take finite real values.
For example: If we assign $x = 1$ *and* $y = 2$ then we get $z = 1 + 2i$ which is a complex number that represents a unique point $(1,2)$ in the complex plane.

Five ways of thinking about a complex number.

Formal (x,y)	Algebraic	Rectangular: $z = x + i\,y$	Exponential: $z = re^{i\theta}$
	Geometric	Cartesian 	Polar 

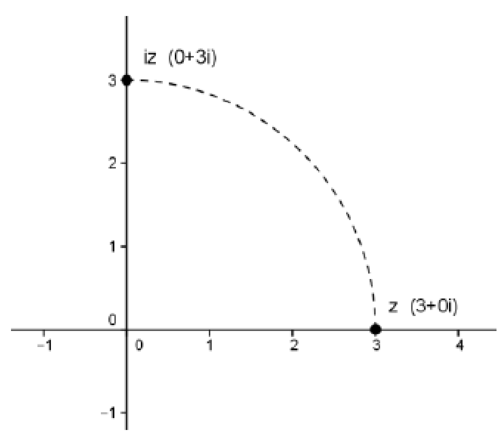
Case 01: Consider x axis as real axis and y-axis as imaginary axis.
Case 02: The ordered pair form of complex number is $(x\ ,y)$.
Case 03: The value of $x=0$ implies $i\,y=(0,y)$ that is a purely imaginary number.
Case 04: The value of $y=0$ implies $z=x=(x,0)$ that is a purely real number.
Case 05: Every real number can be expressed as complex number.
Case06: Geometrically complex number represents a fixed point in the complex plane/Argand Plane/Argand diagram/Gaussian Plane. The number $z=x+i\,y=(x,y)$ by a position vector in the xy plane whose tail is at the origin and whose head is at the point (x,y).
Case 07: Conjugate of a complex number $z=x+i\,y$ is obtained by changing the sign of y and is denoted by the symbol $\bar{z}$ i.e. $\bar{z}=x-iy$ .Geometrically conjugate of a complex number represents the reflection or image of the complex number z about the real axis x.



Historical Notes on complex number:
The geometrical explanation of complex number is given by J.R. Argand and also, he first gives the idea of modulus to represent the value of vector and complex number. The mathematician Leonhard Euler in 1748 introduce the symbol i and its square $i^2=-1$where i is considered as imaginary unit. Gerolamo Cardano named it as Fictitious or sophistic number.German mathematician Carl Friedrich Gauss gave the geometrical interpretation 1832 and Iris mathematician William Rawan Hamilton explained the complex number as Ordered pairs in 1835.

Effect of the symbol i:

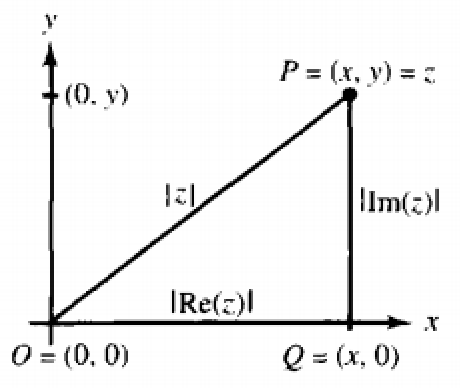
Multiplying a complex number by i induces a 90° rotation about the origin. Each time we multiply by i results in another counterclockwise rotation of 90° about the origin; for example, multiplying by i twice results in a 180° rotation about the origin, and multiplying by i four times results in a full rotation about the origin.



Modulus/Amplitude/Absolute value:

The modulus of a complex of the complex number $z = x + iy$ is a nonnegative real number denoted by $|z| = r \geq 0$ and is defined by the equation $z = \sqrt{x^2 + y^2}$.

The number $|z|$ is the distance between the origin and the point (x, y)in Argand Plane or Gaussian plane. The only complex number with modulus zero is the number 0. The numbers | Re(z) | , | Im(z) |, and | z | are the lengths of the sides of the right triangle OPQ, which is shown in the following figure



Note: Some important identities on complex numbers

1. $\operatorname{Re}(z) = \dfrac{z + \bar{z}}{2}$
2. $\operatorname{Im}(z) = \dfrac{z - \bar{z}}{2i}$
3. $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\overline{z_1}}{\overline{z_2}}$, $z_2 \neq 0$
4. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
5. $\overline{z_1 . z_2} = \overline{z_1} . \overline{z_2}$
6. $\overline{\overline{z}} = z$
7. $z.\bar{z} = |z|^2$

A beautiful application of equation $z.\bar{z} = |z|^2$ is its use in establishing the triangle inequality.

Theorem: The triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$.

Proof:

We have $|z|^2 = z.\bar{z}$

Replacing z by $(z_1 + z_2)$ from the above relation we get

$$|z_1 + z_2|^2 = (z_1 + z_2).\overline{(z_1 + z_2)}$$

$$|z_1 + z_2|^2 = (z_1 + z_2).(\overline{z_1} + \overline{z_2})$$

$$|z_1+z_2|^2 = z_1.\overline{z_1} + z_1.\overline{z_2} + z_2.\overline{z_1} + z_2.\overline{z_2}$$

$$|z_1+z_2|^2 = |z_1|^2 + z_1.\overline{z_2} + z_2.\overline{z_1} + |z_2|^2$$

$$|z_1+z_2|^2 = |z_1|^2 + z_1.\overline{z_2} + \overline{z_1}.\overline{\overline{z_2}} + |z_2|^2$$

$$|z_1+z_2|^2 = |z_1|^2 + z_1.\overline{z_2} + \overline{\left(z_1.\overline{z_2}\right)} + |z_2|^2$$

$$|z_1+z_2|^2 = |z_1|^2 + 2\,\mathrm{Re}\left(z_1.\overline{z_2}\right) + |z_2|^2 \le |z_1|^2 + 2\left|\mathrm{Re}\left(z_1.\overline{z_2}\right)\right| + |z_2|^2$$

$$|z_1+z_2|^2 \le |z_1|^2 + 2\left|z_1.\overline{z_2}\right| + |z_2|^2$$

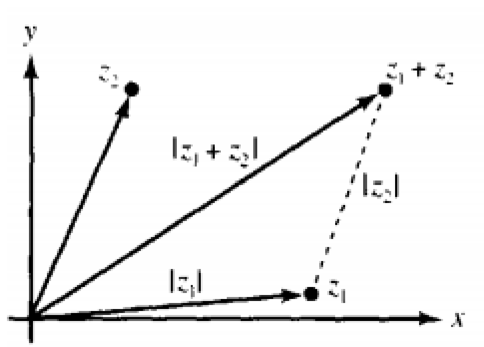
$$|z_1+z_2|^2 \le |z_1|^2 + 2|z_1|.\left|\overline{z_2}\right| + |z_2|^2$$

$$|z_1+z_2|^2 \le \left(|z_1|+|z_2|\right)^2$$

Taking square roots yields the desired inequality.

$$|z_1+z_2| \le |z_1|+|z_2| \qquad \text{(Proved)}$$

Geometrical meaning of this theorem: The triangle inequality states that the sum of the lengths of two sides of a triangle is greater than or equal to the length of the third side.



Note:

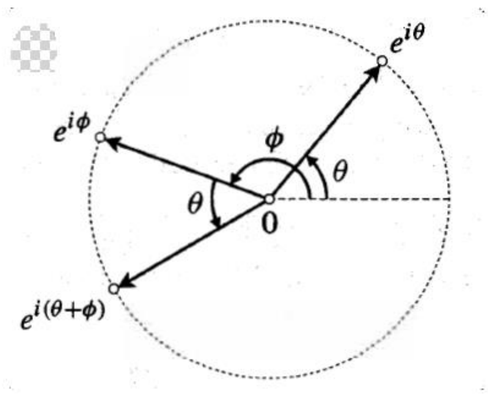
1. The reverse triangle inequality: $|z_1 - z_2| \ge \big||z_1| - |z_2|\big|$
2. The triangle inequality for sum: $\left|\sum_{i=1}^{n} z_i\right| \le \sum_{i=1}^{n} |z_i|$

Argument:

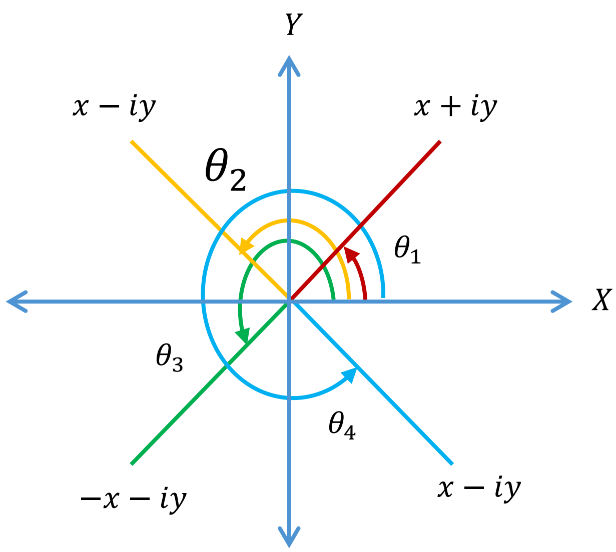
The Argument/Amplitude of the complex number $z = x + i\,y$ is the angle θ such that $x = r\cos\theta,\ y = r\sin\theta$ or $\tan\theta = \dfrac{y}{x}$. A given complex number $z = x + i\,y$ has infinitely many possible arguments. But in the range $-\pi < \theta \le \pi$ every complex number has unique argument called principal argument of that number and is denoted by **Argz.**Generally argument is denoted by arg(z)or amp(z).

Effect of the symbol $e^{i\theta}$:

Multiplication of a complex number $z = e^{i\theta}$ by $e^{i\varphi}$ rotated that number in φ angle.



Formulae for determining argument quadrant-wise:



$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right)$	$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right)$
$\theta_2 = \pi - \tan^{-1}\left(\frac{y}{x}\right)$	$\theta_2 = \pi - \tan^{-1}\left(\frac{y}{x}\right)$
$\theta_3 = \pi + \tan^{-1}\left(\frac{y}{x}\right)$	$\theta_3 = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$
$\theta_4 = 2\pi - \tan^{-1}\left(\frac{y}{x}\right)$	$\theta_4 = -\tan^{-1}\left(\frac{y}{x}\right)$
Argument = arg(z)	Principal Argument = **Arg(z)**

Theorem on argument:

- ❖ $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
- ❖ $\arg(\frac{z_1}{z_2}) = \arg(z_1) - \arg(z_2)$

Euler Formula: The relation $e^{i\theta} = \cos\theta + i\sin\theta = cis\theta$.

Deduction:

We know that the infinite series of trigonometric functions are

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\cdots\cdots + \infty$$

And $$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\cdots\cdots + \infty$$

We have

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots\cdots\cdots + \infty$$

Replacing x by $i\theta$ in the above equation we get,

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} + \cdots\cdots\cdots + \infty$$

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\cdots\cdots + \infty\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\cdots\cdots + \infty\right)$$

$e^{i\theta} = \cos\theta + i\sin\theta$, which is known as **Euler's Formula. (Deduced)**

Roots of a complex number:

A number w is called an n^{th} root of a complex number z that means

$$w = \sqrt[n]{z}$$

By assuming $z = r(\cos\theta + i\sin\theta), z \neq 0$ we get,

$$w = \sqrt[n]{r(\cos\theta + i\sin\theta)}$$

$$w = \left(r(\cos\theta + i\sin\theta)\right)^{\frac{1}{n}}$$

$$w = r^{\frac{1}{n}}(\cos\theta + i\sin\theta)^{\frac{1}{n}}$$

$$w = r^{\frac{1}{n}}(\cos(2k\pi + \theta) + i\sin(2k\pi + \theta))^{\frac{1}{n}} \ , k = 0, 1, 2, \ldots\ldots\ldots, n-1.$$

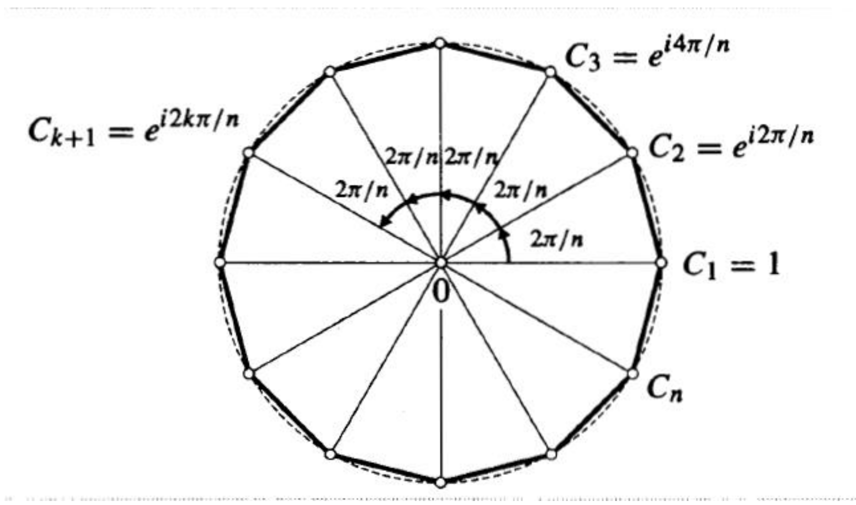
$$w = r^{\frac{1}{n}}\left\{\cos\left(\frac{2k\pi + \theta}{n}\right) + i\sin\left(\frac{2k\pi + \theta}{n}\right)\right\}, k = 0, 1, 2, \ldots\ldots\ldots, n-1.$$

When $\theta = 0 \; and \; r = 1$ then it represents n-th root of unity ,

$$w = \cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right) = e^{\frac{2k\pi i}{n}}, k = 0, 1, 2, \ldots\ldots, n-1.$$

If we let $\omega = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right) = e^{\frac{2\pi i}{n}}$, the n roots are $1, \omega, \omega^2, \omega^3, \ldots\ldots\ldots, \omega^{n-1}$.

Geometrically they represent the n vertices of a regular polygon of n sides inscribed in a circle of radius one with Centre at origin. This circle has the equation $|z| = 1$ and is often called the unit circle.



De Movire's theorem: For all rational values of n , $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

Algebraic and Transcendental numbers:

A number is called algebraic number if it is a solution of a polynomial equation with integer coefficients otherwise it is called transcendental numbers.

Hyperbolic functions:

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \& \sinh x = \frac{e^x - e^{-x}}{2}$$

Note: The roots of the quadratic equation $az^2 + bz + c = 0, \quad a \neq 0$ are $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Binomial Theorem:

$$(x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \cdots\cdots + {}^nC_r x^{n-r} y^r + \cdots\cdots + {}^nC_n x^{n-n} y^n$$

$$= x^n + nx^{n-1}y^1 + \frac{n(n-1)}{2!}x^{n-2}y^2 + \cdots\cdots + \frac{n(n-1)\ldots\ldots\ldots\ldots\ldots(n-r+1)}{r!}x^{n-r}y^r + \cdots\cdots + y^n$$

Worked Out Problems

Find real numbers x and y such that $3x + 2iy - ix + 5y = 7 + 5i$ ..

Solution:

Given complex number is,

$$3x + 2iy - ix + 5y = 7 + 5i$$

$$3x + 5y + (2y - x)i = 7 + 5i$$

Comparing the real part and imaginary part of the complex number from both sides we get,

$$3x + 5y = 7$$

$$2y - x = 5$$

Solving the above equations by calculator we get the values $x = -1$ and $y = 2$. **(As desired)**

Convert the complex number $z = 4e^{-\frac{i\pi}{6}}$ in its Rectangular Form.

Solution:

Given complex number is,

$$z = 4e^{-\frac{i\pi}{6}}$$

$$= 4\left[\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right] \qquad \text{[Using Euler's Formula]}$$

$$= 4\left[\cos\left(\frac{\pi}{6}\right) - i\sin\left(\frac{\pi}{6}\right)\right]$$

$$= 4\left[\frac{\sqrt{3}}{2} - i.\frac{1}{2}\right]$$

$$= 2\sqrt{3} - 2i = x + iy \quad \text{where } x = 2\sqrt{3} \text{ and } y = -2.$$ **(As desired)**

Express $1+i\sqrt{3}$ in the form of $r(\cos\theta + i\sin\theta)$.

Solution:

Given that,

$$1+i\sqrt{3} = r(\cos\theta + i\sin\theta)$$

Comparing the like terms from both-sides we can write

$$r\cos\theta = 1 \;\&\; r\sin\theta = \sqrt{3}$$

Now,

$$r^2\cos^2\theta + r^2\sin^2\theta = 1^2 + \left(\sqrt{3}\right)^2$$

$$r^2\left(\cos^2\theta + \sin^2\theta\right) = 1+3 = 4$$

$$r^2 = 4 \Rightarrow r = 2$$

And

$$\frac{r\sin\theta}{r\cos\theta} = \frac{\sqrt{3}}{1} \Rightarrow \frac{\sin\theta}{\cos\theta} = \sqrt{3} \Rightarrow \tan\theta = \sqrt{3} = \tan\frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$$

Therefore $1+i\sqrt{3} = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ **(As desired)**

Reduce the complex number $\dfrac{(1+2i)^2}{(2+i)^2}$ to the form $A+iB$.

Solution:

Given complex number is,

$$\frac{(1+2i)^2}{(2+i)^2} = \frac{1+2.2i+4i^2}{4+2.2i+i^2}$$

$$= \frac{1+4i-4}{4+4i-1}$$

$$= \frac{-3+4i}{3+4i}$$

$$= \frac{(-3+4i)(3-4i)}{(3+4i)(3-4i)} \qquad \text{[Multiplying denominator and denominator by } (3-4i)\text{]}$$

$$= \frac{(-9+12i+12i+16)}{3^2+4^2}$$

$$= \frac{7+24i}{25}$$

$$= \frac{7}{25} + i\frac{24}{25} = A + iB \quad \text{where } A = \frac{7}{25} \; and \; B = \frac{24}{25}.$$ **(As desired)**

Find the modulus and argument of the complex number $\left(\dfrac{2+i}{3-i}\right)^2$.

Solution:

Given Complex number is,

$$\left(\frac{2+i}{3-i}\right)^2=\frac{(2+i)^2}{(3-i)^2}=\frac{4+4i+i^2}{9-6i+i^2}=\frac{4+4i-1}{9-6i-1}=\frac{3+4i}{8-6i}$$

$$=\frac{3+4i}{8-6i}=\frac{(3+4i)(8+6i)}{(8-6i)(8+6i)}=\frac{24+18i+32i-24}{8^2+6^2}$$

$$=\frac{24+18i+32i-24}{8^2+6^2}$$

$$=\frac{50i}{100}=\frac{i}{2}=0+i.\frac{1}{2}=x+iy \text{ where } x=0 \text{ and } y=\frac{1}{2}.$$

Therefore, the modulus of the given complex number is $=\sqrt{0^2+\left(\frac{1}{2}\right)^2}=\sqrt{0+\frac{1}{4}}=\frac{1}{2}$

And the argument of the given complex number is $=\tan^{-1}\left(\frac{1/2}{0}\right)=\tan^{-1}(\infty)=\frac{\pi}{2}$

Find the modulus and argument of the complex number $z=1+\sin\alpha+i\cos\alpha$.

Solution:

Given complex number is ,

$$z=1+\sin\alpha+i\cos\alpha=x+iy \text{ where } x=1+\sin\alpha, y=\cos\alpha$$

Therefore, the modulus of the given number is $=\sqrt{x^2+y^2}=\sqrt{(1+\sin\alpha)^2+\cos^2\alpha}$

$$=\sqrt{1+2\sin\alpha+\sin^2\alpha+\cos^2\alpha}$$

$$=\sqrt{1+2\sin\alpha+1}$$

$$=\sqrt{2+2\sin\alpha}$$

$$=\sqrt{2(1+\sin\alpha)}$$

$$=\sqrt{2\left(1+\cos\left(\frac{\pi}{2}-\alpha\right)\right)}$$

$$=\sqrt{2.2\cos^2\frac{\left(\frac{\pi}{2}-\alpha\right)}{2}}$$

$$=2\cos\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)$$

And argument of the given number is $=\tan^{-1}\dfrac{y}{x}=\tan^{-1}\dfrac{\cos\alpha}{1+\sin\alpha}=\tan^{-1}\dfrac{\sin\left(\frac{\pi}{2}-\alpha\right)}{1+\cos\left(\frac{\pi}{2}-\alpha\right)}$

$$= \tan^{-1}\frac{2\sin\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)\cos\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)}{2\cos^2\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)}$$

$$= \tan^{-1}\frac{\sin\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)}{\cos\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)} = \tan^{-1}\tan\left(\frac{\pi}{4}-\frac{\alpha}{2}\right) = \left(\frac{\pi}{4}-\frac{\alpha}{2}\right)$$

(As desired)

Express each of the following complex numbers in polar form.

(a) $\boldsymbol{2+2\sqrt{3}\,i}$ (b) $\boldsymbol{-5+5i}$ (c) $\boldsymbol{-\sqrt{6}-\sqrt{2}i}$ (d) $\boldsymbol{-3i}$

Solution: (a): Modulus or absolute value, $r = \left|2+2\sqrt{3}\,i\right| = \sqrt{4+12} = 4$

Amplitude or Argument, $\theta = tan^{-1}(\,2\sqrt{3}/\sqrt{3}) = \pi/3$

Then, $2+2\sqrt{3}\,i = r\,(cos\,\theta + i\,sin\,\theta) = 4\left(cos\frac{\pi}{3} + i\,sin\frac{\pi}{3}\right) = 4e^{\frac{\pi i}{3}}$

(b): Modulus or absolute value, $r = |-5+5i| = \sqrt{25+25} = 5\sqrt{2}$

Amplitude or Argument, $\theta = tan^{-1}\left(\frac{5}{-5}\right) = \pi - tan^{-1}(1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

Then, $-5+5i = r\,(cos\,\theta + i\,sin\,\theta) = 5\sqrt{2}\left(cos\frac{3\pi}{4} + i\,sin\frac{3\pi}{4}\right) = 5\sqrt{2}\,e^{\frac{3\pi i}{4}}$

(c): Modulus or absolute value, $r = \left|-\sqrt{6}-\sqrt{2}i\right| = \sqrt{6+2} = 2\sqrt{2}$

Amplitude or Argument, $\theta = tan^{-1}\left(\frac{-\sqrt{2}}{-\sqrt{6}}\right) = \pi + tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$

Then, $-\sqrt{6}-\sqrt{2}i = r\,(cos\,\theta + i\,sin\,\theta) = 2\sqrt{2}\left(cos\frac{7\pi}{6} + i\,sin\frac{7\pi}{6}\right) = 2\sqrt{2}\,e^{\frac{7\pi i}{6}}$

(d): Modulus or absolute value, $r = |-3i| = \sqrt{0+9} = 3$

Amplitude or Argument, $\theta = tan^{-1}\left(\frac{-3}{0}\right) = \pi + tan^{-1}(\infty) = \pi + \frac{\pi}{2} = \frac{3\pi}{2}$

Then, $-3i = r\,(cos\,\theta + i\,sin\,\theta) = 3\left(cos\frac{3\pi}{2} + i\,sin\frac{3\pi}{2}\right) = 3\,e^{\frac{3\pi i}{2}}$

Prove the identities:

(a) $\boldsymbol{cos\,5\theta = 16\,cos^5\,\theta - 20\,cos^3\,\theta + 5\,cos\,\theta}$

(b) $\boldsymbol{\frac{sin\,5\theta}{sin\,\theta} = 16\,cos^4\,\theta - 12\,cos^2\,\theta + 1}$; **If** $\boldsymbol{\theta \neq 0, \ \pm\pi, \ \pm 2\pi}$**, ….**

Solution:

We have,

$cos\,5\theta + i\,sin\,5\theta = (cos\,\theta + i\,sin\,\theta)^5$

$= cos^5\,\theta + 5\,cos^4\,\theta\,(i\,sin\,\theta) + \frac{5.4}{2!}cos^3\,\theta\,(i\,sin\,\theta)^2 + \frac{5.4.3}{3!}cos^2\,\theta\,(i\,sin\,\theta)^3 + \frac{5.4.3.2}{4!}cos^1\,\theta\,(i\,sin\,\theta)^4 + (i\,sin\,\theta)^5$

$= cos^5\,\theta + 5i\,cos^4\,\theta\,sin\,\theta - 10\,cos^3\,\theta\,sin^2\,\theta - 10i\,cos^2\,\theta\,sin^3\,\theta + 5\,cos\,\theta\,sin^4\,\theta + i\,sin^5\,\theta$

$= cos^5\,\theta - 10\,cos^3\,\theta\,sin^2\,\theta + 5\,cos\,\theta\,sin^4\,\theta + i(5\,cos^4\,\theta\,sin\,\theta - 10\,cos^2\,\theta\,sin^3\,\theta + sin^5\,\theta)$

(a) Comparing Real terms on both sides, we get

$$cos\,5\theta = cos^5\,\theta - 10\,cos^3\,\theta\,sin^2\,\theta + 5\,cos\,\theta\,sin^4\,\theta$$

$$= cos^5\,\theta - 10\,cos^3\,\theta\,(1-cos^2\,\theta) + 5\,cos\,\theta\,(1-cos^2\,\theta)^2$$

$$\therefore\ cos\,5\theta = 16\,cos^5\,\theta - 20\,cos^3\,\theta + 5\,cos\,\theta$$

(b) Comparing imaginary terms on both sides, we get

$$sin\,5\theta = 5\,cos^4\,\theta\,sin\,\theta - 10\,cos^2\,\theta\,sin^3\,\theta + sin^5\,\theta$$

$$\Rightarrow \frac{\sin 5\theta}{\sin \theta} = 5\cos^4\theta - 10\cos^2\theta \sin^2\theta + \sin^4\theta$$
$$= 5\cos^4\theta - 10\cos^2\theta\,(1-\cos^2\theta) + (1-\cos^2\theta)^2$$
$$\frac{\sin 5\theta}{\sin \theta} = 16\cos^4\theta - 12\cos^2\theta + 1;$$

Provided $\sin\theta \neq 0$ i.e. $\theta \neq 0, \pm\pi, \pm 2\pi, \ldots.$

Evaluate each of the following.

(a) $[3(\cos 40^o + i\sin 40^o)][4(\cos 80^o + i\sin 80^o)]$

(b) $\dfrac{(2\,cis\,15^o)^7}{(4\,cis\,45^o)^3}$ (c) $\left(\dfrac{1+\sqrt{3}\,i}{1-\sqrt{3}\,i}\right)^{10}$

Solution: (a) $[3(\cos 40^o + i\sin 40^o)][4(\cos 80^o + i\sin 80^o)]$

$$= [3.e^{40^o i}][4.e^{80^o i}] = 12e^{40^o i + 80^o i} = 12e^{120^o i} = 12(\cos 120^o + i\sin 120^o)$$
$$= 12\{\cos(180^0 - 60^o) + i\sin(180^0 - 60^o)\}$$
$$= 12\{-\cos(60^o) + i\sin(60^o)\} = 12\left(-\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i\right) = -6 + 6\sqrt{3}\,i$$

(b) $$\frac{(2cis15^0)^7}{(4cis45^0)^3} = \frac{(2.e^{15^0 i})^7}{(4.e^{45^0 i})^3} = \frac{2^7(e^{15^0 i})^7}{4^3(e^{45^0 i})^3} = \frac{128.e^{105^0 i}}{64.e^{135^0 i}} = 2.e^{105^0 - 135^0 i} = 2.e^{-30^0 i} = 2\{\cos(-30^0) + i\sin(-30)\}$$

$$= 2\{\cos(30^0) - i\sin(30)\} = 2\left(\frac{\sqrt{3}}{2} - i.\frac{1}{2}\right) = \sqrt{3} - i$$

(c) $\left(\frac{1+\sqrt{3}\,i}{1-\sqrt{3}\,i}\right)^{10} = ?$

Since $|1+\sqrt{3}\,i| = 2$ and $|1-\sqrt{3}\,i| = 2$

$arg(1+\sqrt{3}\,i) = \frac{\pi}{3}$ and $arg(1-\sqrt{3}\,i) = \frac{5\pi}{3}$

$$\therefore \left(\frac{1+\sqrt{3}\,i}{1-\sqrt{3}\,i}\right)^{10} = \left(\frac{2e^{\frac{\pi i}{3}}}{2e^{\frac{5\pi i}{3}}}\right)^{10} = \left(e^{\frac{\pi i}{3} - \frac{5\pi i}{3}}\right)^{10} = \left(e^{-\frac{4\pi i}{3}}\right)^{10} = e^{-\frac{40\pi i}{3}} = e^{-14\pi i + \frac{2\pi i}{3}} = e^{-14\pi i}.e^{\frac{2\pi i}{3}}$$

$$= [\cos(-14\pi) + i\ \sin(-14\pi)]\left[\cos\left(\frac{2\pi}{3}\right) + i\ \sin\left(\frac{2\pi}{3}\right)\right] = (1)\left[-\frac{1}{2} + \frac{\sqrt{3}}{2}\right] = -\frac{1}{2} + \frac{\sqrt{3}}{2}$$

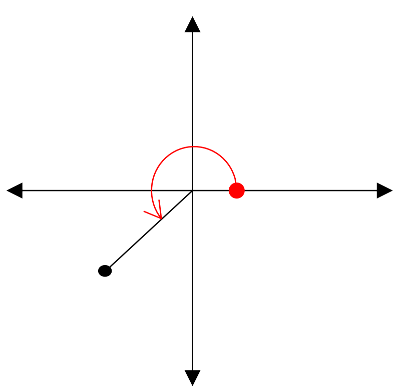
Graph each of the following:

i) $6(\cos 240^0 + i\sin 240)$ ii) $4e^{\frac{3\pi i}{5}}$ iii) $2e^{-\frac{\pi i}{4}}$

Answers:

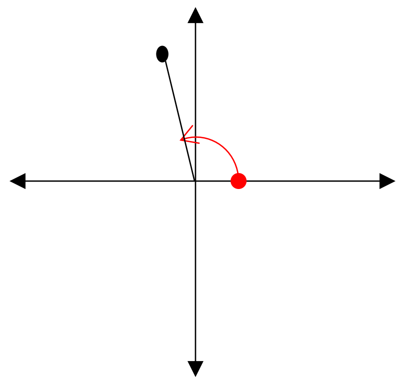
i) Given complex number is $z = 6(\cos 240^0 + i\sin 240) = 6cis240 = 6e^{\frac{4\pi i}{3}}$

Here z represents a complex number with magnitudes 6 and makes 240^0 angle with x-axis in counter clockwise direction.



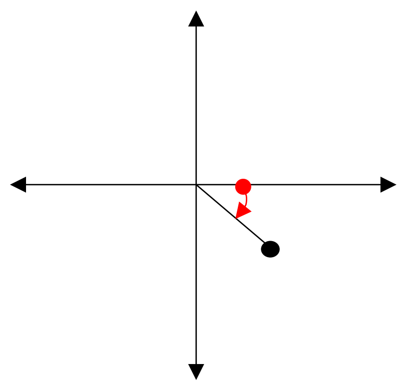
ii) Given complex number is $z = 4e^{\frac{3\pi i}{5}}$

Here z represents a complex number with magnitudes 4 and makes 108^0 angle with x-axis in counter clockwise direction.



iii) Given complex number is $z = 2e^{-\frac{\pi i}{4}}$

Here z represents a complex number with magnitudes 2 and makes 45^0 angle with x-axis in clockwise direction.



 Prove that $\sqrt{3}+\sqrt{2}$ is a algebraic number.

Answers:

a) Assume that $z = \sqrt{3}+\sqrt{2}$.

$$z-\sqrt{2} = \sqrt{3}$$

Squaring both-sides we get,

$$\left(z-\sqrt{2}\right)^2 = 3$$

$$\Rightarrow \ z^2 - 2\sqrt{2}z + 2 = 3$$

$$\Rightarrow \ z^2 - 2\sqrt{2}z - 1 = 0$$

$$\Rightarrow \ z^2 - 1 = 2\sqrt{2}z$$

Again, squaring both-sides we get,

$$\left(z^2-1\right)^2 = \left(2\sqrt{2}z\right)^2$$

$$z^4 - 2z^2 + 1 = 4.2.z^2$$

$$z^4 - 2z^2 + 1 = 8z^2$$

$$z^4 - 10z^2 + 1 = 0$$

This is a polynomial equation with integer coefficients having $\sqrt{3}+\sqrt{2}$ as a root.
Therefore $\sqrt{3}+\sqrt{2}$ is an algebraic number.

(As desired)

Evaluate the value of $\left|e^{iz}\right|$ where $z=6e^{\frac{\pi i}{3}}$.

Solution:

Given that $z=6e^{\frac{\pi i}{3}}$

Here $\left|e^{iz}\right|=\left|e^{i6.e^{\frac{\pi i}{3}}}\right|=\left|e^{6i\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)}\right|$

$$=\left|e^{6i\left(\frac{\sqrt{3}}{2}+i.\frac{1}{2}\right)}\right|=\left|e^{(3\sqrt{3}i-3)}\right|$$

$$=\left|e^{3\sqrt{3}i}.e^{-3}\right|$$

$$=\left|e^{-3}\right|\left|e^{3\sqrt{3}i}\right|$$

$$=e^{-3}\times\left|\cos 3\sqrt{3}+i\cos 3\sqrt{3}\right|$$

$$=e^{-3}\times\sqrt{\cos^2 3\sqrt{3}+\sin^2 3\sqrt{3}}$$

$$=e^{-3}\times\sqrt{1}=e^{-3}$$

(As desired)

Show that the ellipse $|z+3|+|z-3|=10$ can be expressed in rectangular form as $\frac{x^2}{25}+\frac{y^2}{16}=1$ and sketch it.

Solution:

Let $z=x+iy$ where x and y are real.

Given that,

$$|z+3|+|z-3|=10$$

$$|x+iy+3|+|x+iy-3|=10$$

$$|x+3+iy|+|x-3+iy|=10$$

$$\sqrt{(x+3)^2+y^2}+\sqrt{(x-3)^2+y^2}=10$$

$$\sqrt{(x+3)^2+y^2}=10-\sqrt{(x-3)^2+y^2}$$

Squaring both-sides, we get

$$(x+3)^2+y^2=\left\{10-\sqrt{(x-3)^2+y^2}\right\}^2$$

$$(x+3)^2+y^2=100-20\sqrt{(x-3)^2+y^2}+(x-3)^2+y^2$$

$$(x+3)^2-(x-3)^2=100-20\sqrt{(x-3)^2+y^2}$$

$$4x.3=100-20\sqrt{(x-3)^2+y^2}$$

$$12x-100=-20\sqrt{(x-3)^2+y^2}$$

$$3x-25=-5\sqrt{(x-3)^2+y^2}$$

Again, squaring both-sides, we get

$$(3x-25)^2=25\left\{(x-3)^2+y^2\right\}$$

$$9x^2-150x+625=25\left\{x^2-6x+9+y^2\right\}$$

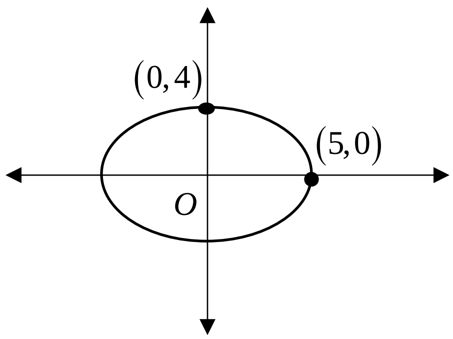
$$9x^2-150x+625=25x^2-150x+225+25y^2$$

$$16x^2+25y^2=400$$

$$\frac{x^2}{25}+\frac{y^2}{16}=1 \text{ which is a rectangular form.}$$

Graph:

Rectangular equation shows that given equation represents an ellipse with Centre at origin and x as major axis and y as minor axis.



Represent graphically the set of values of z for which a) $\left|\frac{z-3}{z+3}\right|=2$ b) $\left|\frac{z-3}{z+3}\right|<2$

Solution:

a)

Given equation is,

$$\left|\frac{z-3}{z+3}\right|=2$$

$$\frac{|z-3|}{|z+3|}=2$$

$$|z-3|=2|z+3|$$

$$|x+iy-3|=2|x+iy+3| \qquad [\because z=x+iy]$$

$$|x-3+iy|=2|x+3+iy|$$

$$\sqrt{(x-3)^2+y^2}=2\sqrt{(x+3)^2+y^2}$$

Squaring both-sides we get ,

$$(x-3)^2+y^2=4\left\{(x+3)^2+y^2\right\}$$

$$x^2-6x+9+y^2=4\left(x^2+6x+9+y^2\right)$$

$$x^2-6x+9+y^2=4x^2+24x+36+4y^2$$

$$3x^2+30x+27+3y^2=0$$

$$x^2+10x+9+y^2=0$$

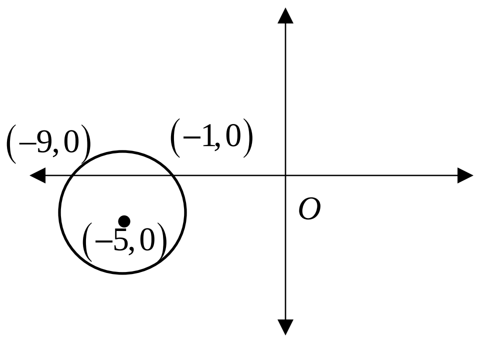
$$x^2+2.x.5+5^2-5^2+9+y^2=0$$

$$(x+5)^2+y^2=16$$

$$(x+5)^2+y^2=4^2$$

$$\{x-(-5)\}^2+(y-0)^2=4^2$$

It represents a circle with Centre $(-5,0)$ and radius 4 units and its graph as follows.



b)

Given equation is,

$$\left|\frac{z-3}{z+3}\right|<2$$

$$\frac{|z-3|}{|z+3|}<2$$

$$|z-3|<2|z+3|$$

$$|x+iy-3|<2|x+iy+3| \qquad [\because z=x+iy]$$

$$|x-3+iy|<2|x+3+iy|$$

$$\sqrt{(x-3)^2+y^2}<2\sqrt{(x+3)^2+y^2}$$

Squaring both-sides we get,

$$(x-3)^2+y^2<4\{(x+3)^2+y^2\}$$

$$x^2-6x+9+y^2<4(x^2+6x+9+y^2)$$

$$x^2-6x+9+y^2<4x^2+24x+36+4y^2$$

$$3x^2+30x+27+3y^2>0$$

$$x^2+10x+9+y^2>0$$

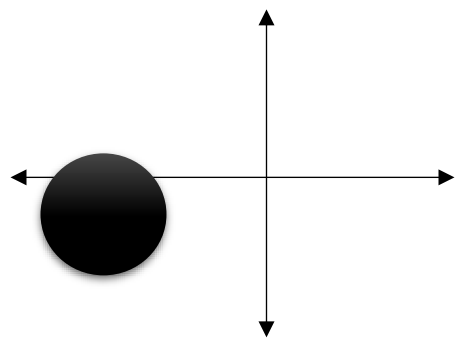
$$x^2+2.x.5+5^2-5^2+9+y^2>0$$

$$(x+5)^2+y^2>16$$

$$(x+5)^2+y^2>4^2$$

$$\{x-(-5)\}^2+(y-0)^2>4^2$$

In convenience of drawing graph considering equality sign it represents a circle with Centre $(-5,0)$ and radius 4 units and its desired graph as follows.



Our given equation represents the set of points in a z-plane outwards the darken region. **(As desired)**

Solve the equations

(i) $z^2+(2i-3)z+5-i=0.$

(ii) $6z^4-25z^3+32z^2+3z-10=0.$

(iii) $z^2(1-z^2)=16.$

Solution: (i) Comparing $z^2+(2i-3)z+5-i=0$ with $az^2+bz+c=0$ We get,

$$a=1,\quad b=2i-3\ ,c=5-i$$

So the solutions are $z=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-(2i-3)\pm\sqrt{(2i-3)^2-4(1)(5-i)}}{2(1)}=\frac{3-2i\pm\sqrt{-15-8i}}{2}$

$=\frac{3-2i\pm(1-4i)}{2}=2-3i$ or $1+i.$

(ii) Let $f(z)=6z^4-25z^3+32z^2+3z-10$; By trial solution we get, $f\left(-\frac{1}{2}\right)=0$;

Therefore, $z=-\frac{1}{2}$ or $2z+1$ is a factor.

Now, $6z^4-25z^3+32z^2+3z-10$

$$=3z^3(2z+1)-14z^2(2z+1)+23z(2z+1)-10(2z+1)$$
$$=(2z+1)(3z^3-14z^2+23z-10)$$

Again let $g(z)=3z^3-14z^2+23z-10$, by trial method, $g(2/3)=0$, so $z=\frac{2}{3}$ or $(3z-2)$ is a factor of $g(z)$.

$$\therefore\ 3z^3-14z^2+23z-10=z^2(3z-2)-4z(3z-2)+5(3z-2)$$
$$=(3z-2)(z^2-4z+5)$$
$$\therefore 6z^4-25z^3+32z^2+3z-10=(2z+1)(3z-2)(z^2-4z+5)=0$$

Hence, $z=-\frac{1}{2},\frac{2}{3}$ and $(z^2-4z+5)=0\Rightarrow z=\frac{4\pm\sqrt{16-20}}{2}=\frac{4\pm\sqrt{-4}}{2}=\frac{4+2i}{2}=2\pm i.$

Thus the solutions are $-\frac{1}{2},\frac{2}{3},\ 2+i,2-i.$

(iii) $z^2(1-z^2)=z^2-z^4=16\Rightarrow z^4-z^2+16=0\Rightarrow z^4+8z^2+16-9z^2=0$

$$\Rightarrow(z^2+4)^2-9z^2=0\ \Rightarrow(z^2+4+3z)(z^2+4-3z)=0$$
$$\Rightarrow\ (z^2+4+3z)=0\ and\ (z^2+4-3z)=0$$
$$\therefore(z^2+4+3z)=0\Rightarrow z=\frac{-3\pm\sqrt{9-16}}{2}=\frac{-3\pm\sqrt{7}i}{2}$$
$$\therefore(z^2+4-3z)=0\Rightarrow z=\frac{3\pm\sqrt{9-16}}{2}=\frac{3\pm\sqrt{7}i}{2}$$

Therefor the roots are: $-\frac{3}{2}\pm\frac{\sqrt{7}}{2}i$ and $\frac{3}{2}\pm\frac{\sqrt{7}}{2}i$

Find all the 8^{th} roots of unity and locate these points in a complex plane. Or Find all values of z for which $z^8 = 1$ and exhibit them graphically. Or solve $z^8 = 1$ for z and sketch them.

Solution:

Here $\quad z^8 = 1$

$$z^8 = 1 + 0.i$$

$$z^8 = \cos 0^0 + i\sin 0^0$$

$$z^8 = \cos\left(2k\pi + 0^0\right) + i\sin\left(2k\pi + 0^0\right) \text{ where } k = 0,1,2,3,\ldots\ldots,7$$

$$z^8 = \cos 2k\pi + i\sin 2k\pi$$

$$z = \left(\cos 2k\pi + i\sin 2k\pi\right)^{\frac{1}{8}} \qquad \text{[taking power } \frac{1}{8} \text{ on both-sides]}$$

$$z = \left(\cos\frac{2k\pi}{8} + i\sin\frac{2k\pi}{8}\right)$$

$$z = \left(\cos\frac{k\pi}{4} + i\sin\frac{k\pi}{4}\right)$$

Putting $k = 0,1,2,3,\ldots\ldots,7$ successively we get the desired roots,

$$z = \left(\cos 0 + i\sin 0\right) = 1$$

$$z = \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1+i)$$

$$z = \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = i$$

$$z = \left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right) = \left(\cos\left(\pi - \frac{\pi}{4}\right) + i\sin\left(\pi - \frac{\pi}{4}\right)\right) = -\cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(-1+i)$$

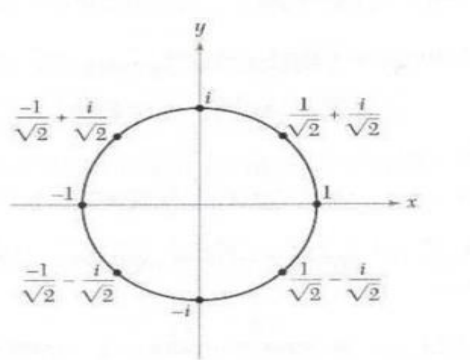
$$z = \left(\cos\pi + i\sin\pi\right) = -1$$

$$z = \left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = -i$$

$$z = \left(\cos\frac{6\pi}{4} + i\sin\frac{6\pi}{4}\right) = \left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = \frac{1}{\sqrt{2}}(-1-i)$$

$$z = \left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right) = \frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1-i)$$

The points are in the complex plane shown as follows



The eight 8th roots of unity.

(As desired)

□□ Find each of the indicated roots $z=(1+i)^{\frac{1}{4}}$.

Solution:

Given complex number is $z=(1+i)^{\frac{1}{4}}$

Also, let, $r\cos\theta=1$ and $r\sin\theta=1$

$$\therefore r^2\cos^2\theta+r^2 sin^2\theta=1^2+1^2=2$$

$$r^2\left(\cos^2\theta+sin^2\theta\right)=2$$

$$r^2=2$$

$$r=\sqrt{2}$$

And

$$\frac{r\sin\theta}{r\cos\theta}=1$$

$$\tan\theta=\tan\frac{\pi}{4}$$

$$\theta=\frac{\pi}{4}$$

Therefore, the polar form of the given complex number is

$$z=\left(\sqrt{2}\cos\frac{\pi}{4}+i\sqrt{2}\sin\frac{\pi}{4}\right)^{\frac{1}{4}}$$

$$z=\left(\sqrt{2}\right)^{\frac{1}{4}}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)^{\frac{1}{4}}$$

$$z=\left(\sqrt{2}\right)^{\frac{1}{4}}\left(\cos\left(2\pi k+\frac{\pi}{4}\right)+i\sin\left(2\pi k+\frac{\pi}{4}\right)\right)^{\frac{1}{4}},k=0,1,2,3$$

$$z=\left(\sqrt{2}\right)^{\frac{1}{4}}\left(\cos(8k+1)\frac{\pi}{4}+i\sin(8k+1)\frac{\pi}{4}\right)^{\frac{1}{4}},k=0,1,2,3$$

$$z=\left(\sqrt{2}\right)^{\frac{1}{4}}\left(\cos(8k+1)\frac{\pi}{16}+i\sin(8k+1)\frac{\pi}{16}\right),k=0,1,2,3$$

□□ Find the all values of $z^5=-32$ and locate these points in a complex plane.

Solution:

$$z^5=-32$$

Here

$$z^5=-32=32(-1)=32(\cos\pi+i\sin\pi)$$

$$z^5=32\left(\cos(2\pi k+\pi)+i\sin(2\pi k+\pi)\right)$$

$$z=\left[32\left(\cos(2\pi k+\pi)+i\sin(2\pi k+\pi)\right)\right]^{\frac{1}{5}}$$

$$z=\left[2^5\left(\cos(2\pi k+\pi)+i\sin(2\pi k+\pi)\right)\right]^{\frac{1}{5}}$$

$$z = 2\left(\cos\left(\frac{2\pi k+\pi}{5}\right)+i\sin\left(\frac{2\pi k+\pi}{5}\right)\right) \qquad \text{[According to de moivre theorem]}$$

Putting k=0,1,2,3,4 we get the desired solutions as follows

$$z = 2\left(\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}\right) = 2e^{\frac{\pi i}{5}}$$

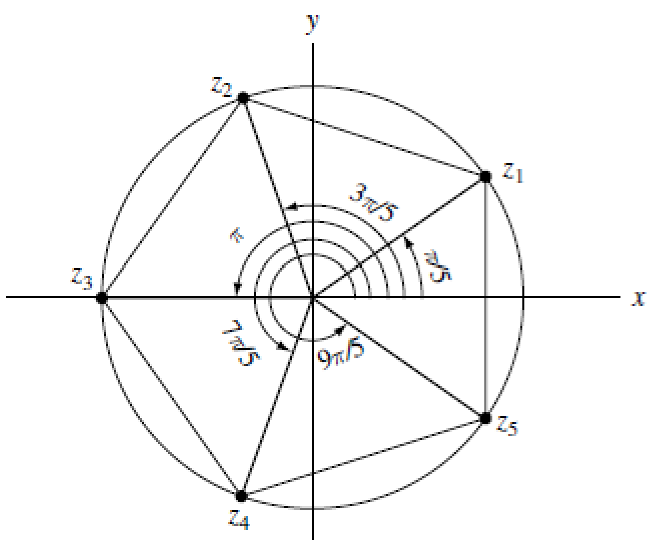
$$z = 2\left(\cos\frac{3\pi}{5}+i\sin\frac{3\pi}{5}\right) = 2e^{\frac{3\pi i}{5}}$$

$$z = 2(\cos\pi + i\sin\pi) = 2e^{\pi i}$$

$$z = 2\left(\cos\frac{7\pi}{5}+i\sin\frac{7\pi}{5}\right) = 2e^{\frac{7\pi i}{5}}$$

$$z = 2\left(\cos\frac{9\pi}{5}+i\sin\frac{9\pi}{5}\right) = 2e^{\frac{9\pi i}{5}}$$

From the calculating roots, it is seen that it represents the vertices of a regular polygon inscribed a circle with radius 2 units and the graph is as follows.



Find all solutions of the equation $\cosh z = 2$.

Solution:

Given equation is $\cosh z = 2$

$$\frac{e^z + e^{-z}}{2} = 2$$

$$e^z + e^{-z} = 4$$

Multiplying by e^z we get,

$$e^{2z} + 1 = 4e^z$$

$$\left(e^z\right)^2 - 4e^z + 1 = 0$$

$$e^z = \frac{4 \pm \sqrt{16-4}}{2}$$

$$e^z = \frac{4 \pm \sqrt{12}}{2}$$

$$e^z = \frac{4 \pm 2\sqrt{3}}{2}$$

$$e^z = 2 \pm \sqrt{3}$$

$$z = \ln\left(2 \pm \sqrt{3}\right) \qquad \textbf{(As desired)}$$

Express $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Solution:

Here

$$(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3\cos^2\theta i\sin\theta + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3$$

$$(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$$

$$(\cos 3\theta + i\sin 3\theta) = \cos^3\theta - 3\cos\theta\sin^2\theta + i\left(3\cos^2\theta\sin\theta - \sin^3\theta\right)$$

Comparing the real part and imaginary part from both sides we get,

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$ and $\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$ **(As desired)**

Mathematical Problems

1. Find the real part and imaginary parts of the complex numbers

i) $z = \dfrac{1}{2+\cos\theta + i\sin\theta}$ ii) z^4 iii) $\dfrac{1}{z^2}$ iv) i^n

2. Reduce the quantities to a real numbers

i) $\dfrac{5i}{(1-i)(2-i)(3-i)}$ ii) $\dfrac{1+2i}{3-4i} + \dfrac{2-i}{5i}$ iii) $(1-i)^4$

3.Compute i) $\dfrac{2+i}{2-i}$ ii) $(1-2i)^4$ iii) $\sin\left(\dfrac{\pi}{4}+i\right)$ iv) $\cos\left(\dfrac{\pi}{3}+i\right)$ v) $\left(5cis20^0\right)\left(3cis40^0\right)$ vi) $\left(2cis50^0\right)^6$

vii) $\dfrac{\left(8cis40^0\right)^3}{\left(2cis60^0\right)^4}$ viii) $\dfrac{\left(3e^{\frac{\pi i}{6}}\right)\left(2e^{-\frac{5\pi i}{4}}\right)\left(6e^{\frac{5\pi i}{3}}\right)}{\left(4e^{\frac{2\pi i}{3}}\right)^2}$ ix) $\left(\dfrac{\sqrt{3}-i}{\sqrt{3}+i}\right)^2\left(\dfrac{1+i}{1-i}\right)^2$

4.Use binomial theorem to expand i) $\left(1+\sqrt{3}i\right)^{2011}$ ii) $\left(1+\sqrt{3}i\right)^{-2011}$

5.Express the following in the form $x+iy$ with $x, y \in R$

i) $\dfrac{i}{1-i} + \dfrac{1-i}{i}$ ii) All the 3^{rd} roots of $(-8i)$ iii) $\left(\dfrac{i+1}{\sqrt{2}}\right)^{1337}$ iv) $(1-i)^3$ v) $\dfrac{1-i}{1+i} - i + 2$

6.Find the exponential form of i) $z = \dfrac{i}{1+i}$ ii) $z = \sqrt{3}+i$ iii) $z = 2-i$

7. Use exponential form to compute i) $\left(1+\sqrt{3}i\right)^{2011}$ ii) $\left(1+\sqrt{3}i\right)^{-2011}$

8. Prove that i) $\displaystyle\sum_{m=0}^{1005}\binom{2011}{2m}(-3)^m = 2^{2010}$ ii) $\displaystyle\sum_{m=0}^{1005}\binom{2011}{2m+1}(-3)^m = 2^{2010}$

10.Show that $\tanh^{-1} z = \dfrac{1}{2}\ln\left(\dfrac{1+z}{1-z}\right)$

11.Put $\left(\sqrt{3}+i\right)^7$ & $\left(-1+i\right)^{100}$ in a Rectangular Form.

12.Write in rectangular form i) $2e^{3\pi i}$ ii) $e^{100\pi i}$ iii) $\sqrt{2}e^{\frac{5\pi i}{4}}$

13. i) Find the polar form of $(1+i)\left(1+i\sqrt{3}\right)$.

ii) Use the result of (i) to find $\cos\left(\frac{7\pi}{12}\right)$ & $\sin\left(\frac{7\pi}{12}\right)$

14.Express $\frac{(1-i)^{23}}{\left(\sqrt{3}-i\right)^{13}}$ in the form $re^{i\theta}$; $r>0, -\pi\le\theta<\pi$

15.Express $5e^{\frac{3\pi i}{4}}+2e^{-\frac{\pi i}{6}}$ in the form $x+iy$ with $x,y\in R$.

16. Find the value of $(1+i)^n+(1-i)^n$

17.Express $\cos 3\varphi, \cos 4\varphi$ *and* $\sin 5\phi$ in terms of $\cos\varphi$ *and* $\sin\phi$.

18.Perform the required calculation and express the answer in the form $a+ib$

i) $\frac{(3-2i)-i(4+5i)}{(1+i)(2+i)(3+i)}$ ii) $\frac{1+2i}{3-4i}-\frac{4-3i}{2-i}$ iii) i^5 iv) $(1+i)^4+(1-i)^4$ v) $\frac{(1+i)^7}{(1-i)^7}$ vi) $\frac{(8\ cis\ 40^o)^3}{(2\ cis\ 60^o)^4}$

vii) $\left(\frac{\sqrt{3}-i}{\sqrt{3}+i}\right)^4\left(\frac{1+i}{1-i}\right)^5$

19.Find the following quantities

i) $|(1+i)(2+i)|$ ii) $\left|\frac{4-3i}{2-i}\right|$ iii) $\left|(1+i)^{50}\right|$ iv) $|z-1|^{50}$ v) $|z\bar{z}|$ where $z=x+iy$

20.Show the equation $2|z|=|z+i|$ describes a circle.

21.Drive the triple angle formula i) $\cos 3\theta=\cos^3\theta-3\cos\theta\sin^2\theta$ ii) $\sin 3\theta=3\cos^2\theta\sin\theta-\sin^3\theta$

22.Find all solutions of $z^2+2z+(1-i)=0$

23.Sketch the following sets in a complex plane.

(a) $\{z\in\mathbb{C}: |z-1+i|=2\}$.

(b) $\{z\in\mathbb{C}: |z-1+i|\le 2\}$.

(c) $\{z\in\mathbb{C}: \operatorname{Re}(z+2-2i)=3\}$.

(d) $\{z\in\mathbb{C}: |z-i|+|z+i|=3\}$.

(e) $\{z\in\mathbb{C}: |z|=|z+1|\}$.

24.Find all solutions to the following equations

(a) $z^6=1$.

(b) $z^4=-16$.

(c) $z^6=-9$.

(d) $z^6-z^3-2=0$.

26.Use the quadratic formula to solve the following equations. Put your answers in standard form:

(a) $z^2 + 25 = 0$.

(b) $2z^2 + 2z + 5 = 0$.

(c) $5z^2 + 4z + 1 = 0$.

(d) $z^2 - z = 1$.

(e) $z^2 = 2z$.

27. If $|z| = 1$ compute $|z| = |1+z|^2 + |1-z|^2$

28.Solve the following equations for Z:

(a) $|z| - z = 1 + i$; (b) $|z| + z = 1 + i$. (c) $z - |z| = 1 + i$.

29. If $|z| = 1$ and $z \neq 1$ then show that $\frac{1+z}{1-z} = ib$ for some $b \in R$.

30.Find reciprocal of $z = 2 - 3i$

31.Write the given number in the form $a + ib$

(a) $2i^3 - 3i^2 + 5i$

(b) $3i^5 - i^4 + 7i^3 - 10i^2 - 9$

(c) $\frac{5}{i} + \frac{2}{i^3} - \frac{20}{i^{18}}$

(d) $2i^6 + \left(\frac{2}{-i}\right)^3 + 5i^{-5} - 12i$

32.Find the fourth root of $z = 1 + i$.

33.Let $\frac{x - iy}{x + iy} = a + ib$ .Prove that $a^2 + b^2 = 1$

34.Show that (a) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ (b) $\cos 4\theta = 8\sin^4\theta - 8\sin^2\theta + 1$

35.Solve: (a) $z^5 - 2z^4 - z^3 + 6z - 4 = 0$. (b) $z^4 + z^2 + 1 = 0$.

36.Find the modulus and argument of the complex number $z = 1 + \cos\alpha + i\sin\alpha$ **.**

37.Prove that a) $\sqrt[3]{4} - 2i$ b) $\sqrt[3]{2} + \sqrt{3}$ and c) $2 - \sqrt{2}i$ are algebraic numbers.

38. Show that $2 + i = \sqrt{5}e^{i\tan^{-1}(1/2)}$

39. Graph each of the following and express in rectangular form.

i) $6\left(\cos 135^0 + i\sin 135^0\right)$ ii) $4cis315^0$ iii) $5e^{\frac{7\pi i}{6}}$ iv) $3e^{-\frac{2\pi i}{3}}$

40. Express each of the following complex numbers in polar form.

i) $\sqrt{2} + 2\sqrt{2}i$ ii) $\frac{\sqrt{3}}{2} - \frac{3}{2}i$

41. Find real numbers x and y such that $2x-3iy+4ix-2y-5-10i=(x+y+2)-(y-x+3)i$.

42. Find $|w|^2$ in terms of x and y where $w=3iz-z^2$ and $z=x+iy$.

43.Prove that $|z+2i|+|z-2i|=6$ represents an ellipse

44.Prove that $|z-3|-|z+3|=4$ represents a hyperbola

45.Prove that $|z-1|=|z+i|$ represents a straight line passing through the origin whose slope is -1.

46.Prove that $|z-2i|=6$ represents a circle whose Centre is at (0,2) and radius is 6 units.

47.Show that in the Argand's plane a variable point z subject to the condition $|z+1|+|z-1|=3$ describes an ellipse.

48.Describe geometrically the set of points z satisfying the following conditions:

i) $|z-4|\geq|z|$ ii) $|z-i|\leq|z+i|$ iii) $1<|z-2i|<2$ iv) $1<|z+i|\leq 2$

49.Sketch the region in z plane represented by the points:

i. $\operatorname{Re} z(\bar{z}-1)=2$ ii. $\operatorname{Im}(z)=1$ iii. $Re(z^2)>1$ iv. $Re\left(\frac{1}{z}\right)<\frac{1}{2}$

50. Find all roots of i) $z^4+z^2+1=0$ ii) $(1+z)^5=(1-z)^5$